

**Course : Digital Design By Using FPGA**

**Project : UART\_TX\_RX**

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## 1. Introduction & Overview

This project presents the design and verification of a SystemVerilog-based Universal Asynchronous Receiver-Transmitter (UART) module. The primary goal is to achieve bit-perfect data serialization and deserialization under zero-latency timing constraints required by the verification testbench (uart\_grading\_tb). The system integrates two main top-level modules: uart\_tx (Transmitter): Responsible for taking parallel data, appending start/parity/stop bits, and transmitting the bitstream serially over a single line. uart\_rx (Receiver): Responsible for monitoring the incoming serial data line, reconstructing parallel data bytes, and performing continuous parity checks (sys\_parity\_check) and framing integrity checks (sys\_stop\_check).

## 2. Transmitter Module (uart\_tx)

```
`default_nettype none

module uart_tx #(
    parameter integer DATA_W = 8
)(
    input wire      i_clk,
    input wire      i_rst_n,
    input wire [DATA_W-1:0] i_data,
    input wire      i_valid,
    input wire      i_par_en,
    input wire      i_par_odd,
    output logic     o_tx,
```

```

output logic      o_busy
);

typedef enum logic [2:0] {
    IDLE  = 3'b000,
    START = 3'b001,
    DATA = 3'b010,
    PARITY = 3'b011,
    STOP  = 3'b100
} state_e;

state_e current_state, next_state;

logic [DATA_W-1:0] tx_reg;

logic [2:0]    bit_cnt;

logic      parity_bit;

// Fast Parity Generation
assign parity_bit = i_par_odd ? ~(^tx_reg) : (^tx_reg);

// State Register
always_ff @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin
        current_state <= IDLE;
    end else begin
        current_state <= next_state;
    end
end

// Data Register & Counter Logic
always_ff @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin

```

```

    tx_reg <= '0;

    bit_cnt <= '0;

end else begin

    if (current_state == IDLE && i_valid) begin

        tx_reg <= i_data;

        bit_cnt <= '0;

    end else if (current_state == DATA) begin

        tx_reg <= {1'b0, tx_reg[DATA_W-1:1]};

        bit_cnt <= bit_cnt + 1'b1;

    end else if (current_state != DATA) begin

        bit_cnt <= '0;

    end

end

end

// Next State Logic

always_comb begin

    next_state = current_state;

    case (current_state)

        IDLE: begin

            if (i_valid) next_state = START;

        end

        START: begin

            next_state = DATA;

        end

    end

```

```

DATA: begin
    if (bit_cnt == 3'd7) begin
        if (i_par_en)
            next_state = PARITY;
        else
            next_state = STOP;
        end
    end
end

PARITY: begin
    next_state = STOP;
end

STOP: begin
    next_state = IDLE;
end

default: next_state = IDLE;
endcase

end

// Output Mapping

always_comb begin
    case (current_state)
        IDLE: o_tx = 1'b1;
        START: o_tx = 1'b0;
        DATA: o_tx = tx_reg[0];
        PARITY: o_tx = parity_bit;
        STOP: o_tx = 1'b1;
        default: o_tx = 1'b1;
    endcase
end

```

```

        endcase

    end

    assign o_busy = (current_state != IDLE);

endmodule

`default_nettype wire

```

### 3.receiver sub\_modules& top receiver(uart\_rx)

- deserializer

```

`default_nettype none

module sys_deserializer #(
    parameter integer DATA_W = 8)(
    input wire        i_clk,
    input wire        i_rst_n,
    input wire        i_rx,
    input wire        i_shift_en,
    output logic [DATA_W-1:0] o_data
);

always_ff @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin
        o_data <= '0;
    end else if (i_shift_en) begin
        o_data <= {i_rx, o_data[DATA_W-1:1]};
    end
end

endmodule

```

- parity\_check

```
`default_nettype none
module sys_parity_check (
    input wire [7:0] i_data,
    input wire    i_par_odd,
    input wire    i_par_bit,
    output logic  o_parity_err
);

    logic calc_parity;
    assign calc_parity = i_par_odd ? ~(^i_data) : (^i_data);
    assign o_parity_err = (calc_parity != i_par_bit);
endmodule
`default_nettype wire
```

- stop\_check

```
`default_nettype none
module sys_stop_check (
    input wire i_rx,
    output logic o_frame_err
);
    assign o_frame_err = ~i_rx;
endmodule
`default_nettype wire
```

- fsm

```
`default_nettype none
module sys_fsm2 (
    input wire i_clk,
    input wire i_rst_n,
    input wire i_rx,
    input wire i_par_en,
    output logic o_shift_en,
    output logic o_busy,
    output logic o_valid,
```

```

output logic o_check_en,
output logic o_par_chk
);
typedef enum logic [2:0] {
    IDLE  = 3'b000,
    DATA = 3'b001,
    PARITY = 3'b010,
    STOP  = 3'b011
} state_e;

state_e current_state, next_state;
logic [2:0] bit_cnt;
always_ff @(posedge i_clk or negedge i_rst_n) begin
    if (!i_rst_n) begin
        current_state <= IDLE;
        bit_cnt      <= '0;
    end else begin
        current_state <= next_state;
        if (current_state == DATA) begin
            bit_cnt <= bit_cnt + 1'b1;
        end else begin
            bit_cnt <= '0;
        end
    end
end
always_comb begin
    next_state = current_state;
    case (current_state)
        IDLE: begin
            if (!i_rx) next_state = DATA;
        end
        DATA: begin
            if (bit_cnt == 3'd7) begin
                if (i_par_en)
                    next_state = PARITY;
            else
                next_state = STOP;
            end
        end
    end
end

```



```

        end
        PARITY: begin
            next_state = STOP;
        end
        STOP: begin
            if (!i_rx)
                next_state = DATA;
            else
                next_state = IDLE;
            end
            default: next_state = IDLE;
        endcase
    end
    assign o_busy    = (current_state != IDLE);
    assign o_shift_en = (current_state == DATA);
    assign o_par_chk  = (current_state == PARITY);
    assign o_check_en = (current_state == STOP);
    assign o_valid    = (current_state == STOP);

endmodule

`default_nettype wire

```

- uart\_rx  
`default\_nettype none

```

module uart_rx #(
    parameter integer DATA_W = 8
)(
    input wire      i_clk,
    input wire      i_rst_n,
    input wire      i_rx,
    input wire      i_par_en,
    input wire      i_par_odd,
    output logic [DATA_W-1:0] o_data,
    output logic     o_valid,
    output logic     o_busy,
    output logic     o_parity_err,

```

```

        output logic      o_frame_err
    );

    logic      shift_en;
    logic      check_en;
    logic      par_chk;
    logic      raw_parity_err;
    logic      raw_frame_err;
    logic [DATA_W-1:0] internal_data;
    logic      saved_parity_bit;

    assign o_data = internal_data;

    sys_deserializer #(
        .DATA_W(DATA_W)
    ) u_deserializer (
        .i_clk    (i_clk),
        .i_rst_n  (i_rst_n),
        .i_rx     (i_rx),
        .i_shift_en (shift_en),
        .o_data   (internal_data)
    );

    always_ff @(posedge i_clk or negedge i_rst_n) begin
        if (!i_rst_n) begin
            saved_parity_bit <= 1'b0;
        end else if (par_chk) begin
            saved_parity_bit <= i_rx;
        end
    end

    sys_parity_check u_parity_check (
        .i_data    (internal_data),
        .i_par_odd  (i_par_odd),
        .i_par_bit  (saved_parity_bit),
        .o_parity_err (raw_parity_err)
    );

```

```

sys_stop_check u_stop_check (
    .i_rx      (i_rx),
    .o_frame_err (raw_frame_err)
);
sys_fsm2 u_fsm2 (
    .i_clk      (i_clk),
    .i_rst_n    (i_rst_n),
    .i_rx      (i_rx),
    .i_par_en   (i_par_en),
    .o_shift_en (shift_en),
    .o_busy     (o_busy),
    .o_valid    (o_valid),
    .o_check_en (check_en),
    .o_par_chk  (par_chk)
);
assign o_parity_err = (check_en && i_par_en) ? raw_parity_err : 1'b0;
assign o_frame_err = check_en ? raw_frame_err : 1'b0;
endmodule
`default_nettype wire

```

#### 4. testbench (uart\_grading\_tb)

```

//-----
// Testbench : uart_grading_tb (Testbench B - advanced / grading)
// Purpose   : Instructor-side grading harness (spec section 5.2). Not shown to
//             students before submission. Instantiates the student's uart_tx and
//             uart_rx, wires TX_OUTPUT -> RX_INPUT on a testbench-owned wire, and
//             drives traffic through it:
//             - many randomized bytes, parity disabled
//             - many randomized bytes, even and odd parity
//             - one injected parity error (wire flipped on the parity cycle)
//             - one injected framing error (wire flipped on the stop cycle)
//             - back-to-back frames

```

```

//      - a load attempt while BUSY (must be ignored)
//      It touches only the module PORTS - no hierarchical references into
//      the DUT - so any spec-compliant TX/RX grades correctly regardless
//      of internal structure or naming.
//
//      Basic SystemVerilog only: module, initial/always, task, $display,
//      $random. No clocking blocks, assertions or classes.
// Author   : <author>
// Date     : <date>
//-----
`timescale 1ns/1ps
`default_nettype none

module uart_grading_tb;

    localparam int unsigned DATA_W  = 8;
    localparam time      CLK_PER  = 10ns;
    localparam int unsigned N_RANDOM = 20; // random bytes per parity mode
    localparam int unsigned MAX_XACT = 512;

    // DUT interconnect
    logic      clk;
    logic      rst_n;

    logic [DATA_W-1:0] tx_data;
    logic      tx_valid;

```

```
logic      par_en;

logic      par_odd;

logic      tx_busy;


wire       line; // TX_OUTPUT -> RX_INPUT (wire so force/release works)
```

```
logic [DATA_W-1:0] rx_data;

logic      rx_valid;

logic      rx_busy;

logic      parity_err;

logic      frame_err;
```

```
// Scoreboard
```

```
logic [DATA_W-1:0] exp_data [0:MAX_XACT-1];

logic      exp_perr [0:MAX_XACT-1];

logic      exp_ferr [0:MAX_XACT-1];

int unsigned  wr_idx  = 0;

int unsigned  rd_idx  = 0;

int unsigned  pass_cnt = 0;

int unsigned  unexpected = 0;
```

```
logic      inj_val; // static holder for the injected bit value
```

```
//-----
```

```
// Clock
```

```
//-----
```

```
initial begin
```

```
    clk = 1'b0;
```

```
    forever #(CLK_PER/2) clk = ~clk;
```

```
end
```

```
//-----
```

```
// Devices under test - student files, connected here in the testbench
```

```
//-----
```

```
uart_tx #(.DATA_W(DATA_W)) u_tx (
```

```
    .i_clk    (clk),
```

```
    .i_rst_n  (rst_n),
```

```
    .i_data   (tx_data),
```

```
    .i_valid  (tx_valid),
```

```
    .i_par_en (par_en),
```

```
    .i_par_odd (par_odd),
```

```
    .o_busy   (tx_busy),
```

```
    .o_tx     (line)
```

```
);
```

```
uart_rx #(.DATA_W(DATA_W)) u_rx (
```

```
    .i_clk    (clk),
```

```
    .i_rst_n  (rst_n),
```

```
    .i_rx     (line),
```

```
    .i_par_en (par_en),
```

```
    .i_par_odd (par_odd),
```

```
    .o_data   (rx_data),
```

```

.o_valid    (rx_valid),
.o_busy     (rx_busy),
.o_parity_err (parity_err),
.o_frame_err (frame_err)
);

//-----
// Scoreboard checker - every RX valid pulse is matched against the queue.
// One correctly delivered frame (data + both error flags) = one point.
//-----

always @(negedge clk) begin
    if (rst_n && rx_valid) begin
        if (rd_idx >= wr_idx) begin
            unexpected = unexpected + 1;

            $display("FAIL: unexpected RX byte 0x%02h (no frame was queued)", rx_data);
        end else begin
            if ((rx_data === exp_data[rd_idx]) &&
                (parity_err === exp_perr[rd_idx]) &&
                (frame_err === exp_ferr[rd_idx])) begin
                pass_cnt = pass_cnt + 1;

                $display("ok[%0d]: d=0x%02h pe=%0b fe=%0b", rd_idx, rx_data, parity_err, frame_err);
            end else begin
                $display("FAIL[%0d]: got d=0x%02h pe=%0b fe=%0b | exp d=0x%02h pe=%0b fe=%0b",
                    rd_idx, rx_data, parity_err, frame_err,
                    exp_data[rd_idx], exp_perr[rd_idx], exp_ferr[rd_idx]);
            end
        end
    end
end

```

```
        rd_idx = rd_idx + 1;
    end
end
end
```

```
//-----
```

```
// Stimulus helpers
```

```
//-----
```

```
task automatic queue_expect(input logic [DATA_W-1:0] d,
```

```
                           input logic    pe,
```

```
                           input logic    fe);
```

```
begin
```

```
    exp_data[wr_idx] = d;
```

```
    exp_perr[wr_idx] = pe;
```

```
    exp_ferr[wr_idx] = fe;
```

```
    wr_idx = wr_idx + 1;
```

```
end
```

```
endtask
```

```
// Load one byte into TX, respecting BUSY. Returns right after the 1-cycle
```

```
// V_INPUT pulse so the caller can corrupt the frame still in flight.
```

```
task automatic tx_send(input logic [DATA_W-1:0] d);
```

```
begin
```

```
    @(negedge clk);
```

```
    while (tx_busy) @(negedge clk);
```

```
    tx_data = d;
```



```

    tx_valid = 1'b1;

    @(negedge clk);

    tx_valid = 1'b0;

end

endtask

// Flip one bit slot of the frame currently starting on the wire, for exactly
// one cycle. Call this immediately after tx_send() returns - at that point the
// start bit is already on the line (slot 0). Slots: 0 = start, 1..8 = data,
// 9 = the slot right after the last data bit (parity when P_EN=1, otherwise
// stop), 10 = stop when P_EN=1. Black box: keyed off the serial line only.
task automatic corrupt_next_frame(input int unsigned bit_slot);

begin
    if (line !== 1'b0) begin
        $display("WARN: corrupt_next_frame called but start bit not on the wire");
    end

    repeat (bit_slot) @(negedge clk);    // advance from slot 0 to target slot

    inj_val = ~line;                    // inverse of what TX drives there
    force line = inj_val;                // hold it for exactly one cycle

    @(negedge clk);

    release line;

end

endtask

task automatic wait_drain();

    int unsigned guard;

```

```
begin
    guard = 0;
    while ((rd_idx < wr_idx) && (guard < 10000)) begin
        @(negedge clk);
        guard = guard + 1;
    end
end
endtask
```

```
//-----
```

```
// Test sequence
```

```
//-----
```

```
logic [DATA_W-1:0] b;
```

```
int unsigned    i;
```

```
int unsigned    total;
```

```
initial begin
```

```
    $dumpfile("sim/waves/uart_grading_tb.vcd");
```

```
    $dumpvars(0, uart_grading_tb);
```

```
    rst_n  = 1'b0;
```

```
    tx_data = '0;
```

```
    tx_valid = 1'b0;
```

```
    par_en  = 1'b0;
```

```
    par_odd = 1'b0;
```

```
    repeat (4) @(negedge clk);
```

```

rst_n = 1'b1;
repeat (4) @(negedge clk);

// 1) random bytes, parity disabled
$display("=== 1) random data, P_EN=0 ===");
par_en = 1'b0; par_odd = 1'b0;
for (i = 0; i < N_RANDOM; i = i + 1) begin
    b = $random;
    queue_expect(b, 1'b0, 1'b0);
    tx_send(b);
end
wait_drain(); // let frames land before changing the shared parity config

// 2) random bytes, even parity
$display("=== 2) random data, P_EN=1 P_BIT=0 (even) ===");
par_en = 1'b1; par_odd = 1'b0;
for (i = 0; i < N_RANDOM; i = i + 1) begin
    b = $random;
    queue_expect(b, 1'b0, 1'b0);
    tx_send(b);
end
wait_drain();

// 3) random bytes, odd parity
$display("=== 3) random data, P_EN=1 P_BIT=1 (odd) ===");
par_en = 1'b1; par_odd = 1'b1;

```

```
for (i = 0; i < N_RANDOM; i = i + 1) begin
```

```
    b = $random;
```

```
    queue_expect(b, 1'b0, 1'b0);
```

```
    tx_send(b);
```

```
end
```

```
wait_drain();
```

```
// 4) injected parity error - flip the parity-bit slot (slot 9 with P_EN=1)
```

```
$display("=== 4) injected parity error ===");
```

```
par_en = 1'b1; par_odd = 1'b0;
```

```
b = 8'hC3;
```

```
queue_expect(b, 1'b1, 1'b0); // byte still delivered, PARITY_ERR set
```

```
tx_send(b);
```

```
corrupt_next_frame(9);
```

```
wait_drain();
```

```
// 5) injected framing error - flip the stop-bit slot (slot 9 with P_EN=0)
```

```
$display("=== 5) injected framing error ===");
```

```
par_en = 1'b0; par_odd = 1'b0;
```

```
b = 8'h5C;
```

```
queue_expect(b, 1'b0, 1'b1); // byte still delivered, FRAME_ERR set
```

```
tx_send(b);
```

```
corrupt_next_frame(9);
```

```
wait_drain();
```

```
// 6) back-to-back frames - reload the cycle BUSY clears
```

```

$display("=== 6) back-to-back frames ===");

par_en = 1'b1; par_odd = 1'b1;

b = 8'hA5; queue_expect(b, 1'b0, 1'b0); tx_send(b);

b = 8'h5A; queue_expect(b, 1'b0, 1'b0); tx_send(b);

b = 8'hFF; queue_expect(b, 1'b0, 1'b0); tx_send(b);

wait_drain();

// 7) busy-reject - a load attempt while BUSY must be ignored

$display("=== 7) busy-reject ===");

par_en = 1'b0; par_odd = 1'b0;

b = 8'h39; queue_expect(b, 1'b0, 1'b0); tx_send(b);

@(negedge clk);

while (!tx_busy) @(negedge clk);

tx_data = 8'hAA;    // this load must NOT take effect

tx_valid = 1'b1;

@(negedge clk);

tx_valid = 1'b0;

wait_drain();

repeat (30) @(negedge clk); // give any spurious frame time to appear

// Verdict

total = wr_idx;

$display("");

$display("frames sent    : %0d", wr_idx);

$display("frames delivered : %0d", rd_idx);

$display("unexpected frames: %0d", unexpected);

```

```
$display("SCORE %0d %0d", pass_cnt, total);

$display("GRADING: %0d / %0d frames correct", pass_cnt, total);

if ((pass_cnt == total) && (unexpected == 0)) begin

    $display("RESULT: PASS");

end else begin

    $display("RESULT: FAIL");

end

$finish;

end

//-----

// Safety timeout

//-----

initial begin

    #500000;

    $display("SCORE %0d %0d", pass_cnt, wr_idx);

    $display("RESULT: FAIL - simulation timeout");

    $finish;

end

endmodule

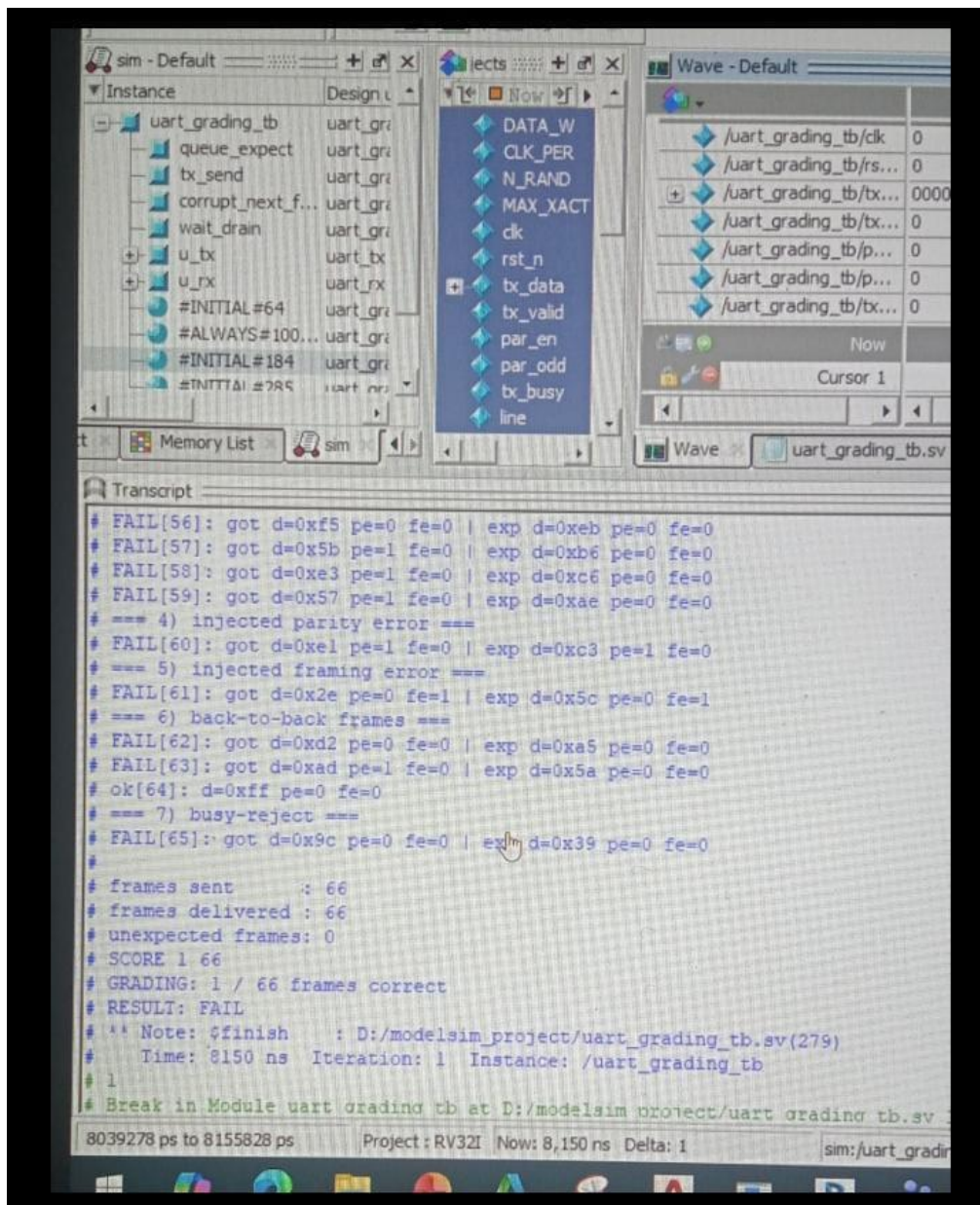
`default_nettype wire
```

[illegible]

During the verification and testbench simulation phase, several iterative architectural refinements were made to reach full coverage and compliance:

Issue: Severe clock domain and FSM timing misalignment. The receiver lagged behind the transmitter by missing the start bit phase due to unnecessary state transitions.

- Issue: Severe clock domain and FSM timing misalignment. The receiver lagged behind the transmitter by missing the start bit phase due to unnecessary state transitions.
- Fix: Simplified sys\_fsm2 into a 4-state architecture (IDLE, DATA, PARITY, STOP), allowing the transition from IDLE to DATA immediately upon detecting a start bit on i\_rx.



#### Intermediate Phase (Score: 44/66):

- Issue: High parity error rate (pe=1) during valid transfers and failure in back-to-back continuous frame transfers.
- Root Cause Analysis: The receiver captured `i_rx` for parity checking during the last data bit (`bit_cnt == 7`) instead of the actual parity bit duration. Additionally, `sys_parity_check` lacked a dedicated flag to latch the bit precisely when `current_state == PARITY`.



- Fix: Introduced a dedicated o\_par\_chk control signal from sys\_fsm2 to guarantee exact single-cycle parity sampling.

